Charbel Fayad

Software Integration Engineer and Scrum Master

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Experience

Software Integration Engineer, Murex — Jun 2024 – Present

- Architect API-led, event-driven integrations on MuleSoft Anypoint connecting 20+ cloud and on-premise systems, real-time and async over ActiveMQ and RabbitMQ.
- Apply enterprise integration patterns (canonical data models, content-based routing, API gateways) and design for high availability with clustering and active-passive failover.
- Built from scratch an Anypoint management-as-code tool that versions API contracts, specifications, instances and gateways and deploys them through CI/CD.
- Built an ActiveMQ observability tool (React, PostgreSQL, Docker) for queues, message tracing and audits.
- Design REST and SOAP APIs from business processes and stakeholder requirements; keep them reliable with Log4j structured logging and Splunk monitoring.
- Maintain Groovy Jenkins pipelines for Maven and Dockerized Node.js: Nexus, Harbor, Black Duck, Prisma Cloud.
- **Scrum Master (SAFe)**: run the ceremonies, coordinate cross-team dependencies, manage risk, present deliveries.
- **Support Coordinator**: triage support requests, analyze recurring issues, coordinate fixes across the team.

Projects

SD-WAN and SASE orchestrator — 2026

Go · gRPC · KVM · Kubernetes · Kafka · Flink · ClickHouse · AWS

- Architected B2B2B multi-tenant orchestration on a custom Linux NFV infrastructure over KVM, with near-zero-touch edge provisioning and service-chained VNFs for SD-WAN, firewalling and inline IDS/IPS.
- Architected the telemetry services: thousands of IoT edge devices stream through Kafka into Flink for real-time processing and ClickHouse for analytics, replicated across availability zones.
- Built Go edge agents that report to regional control planes over gRPC, on AWS and Kubernetes, with the data plane at the edge or in the cloud.

Apps — 2022 – 2026

- **Bishara**: Arabic scripture on the iOS Lock Screen, widgets and wallpapers. Built solo in React Native and Expo with native widget extensions; offline, no accounts.
- **MulePrep**: MuleSoft certification practice engine, React on Cloudflare's edge.
- **Petale**: Bouquet composer in React and Vite that serializes arrangements into shareable links.
- **Rukhsa**: Multilingual PWA for Lebanese driving-theory exams.
- **Socrat**: Open-source WebRTC platform on Next.js for virtual Socratic dialogue, with real-time participation and attendance analytics (2024).
- **Earlier**: Tourify (Ionic, Angular, PHP), a dollar-rate API (FastAPI, Flutter), Smoll (SolidJS), Mandelbrot (C#).

Research

Machine Learning Research, Lebanese American University — Mar 2022 – Jun 2025

Python · TensorFlow · Keras · DEAP · NetworkX

- Genetic algorithm evolving on a dynamic scale-free (Barabási–Albert) network, where authority nodes balance exploration and exploitation.
- Neural network ensemble embedded in a scale-free graph, weighted by node centrality and pruned through its power-law degree distribution.
- Adaptive-topology neural networks that rewire low-importance connections during training; genetic programming for federated learning on binary neural networks.

Publications

- Fayad, C., & Eid, C. (2026). *A Scale-Free Network for Modeling a Homogeneous Ensemble*. Authorea preprint.
- Fayad, C. (2025). *Adaptive Neural Topologies: A Dynamic Approach to Learning Network Structures during Training*. TechRxiv preprint.
- Hoblos, Y., & Fayad, C. (2025). *A Scale-Free Network-Based Genetic Algorithm with Balanced Exploration and Exploitation*. engrXiv preprint.

Education

Lebanese American University, Byblos — Sep 2021 – Jun 2024

- BS Computer Science, Honors Program. CGPA 3.97, Major GPA 4.0, High Distinction.
- **Founder**: Software Engineering Club (workshops and technical seminars) and Computer Science Club.
- **Lab Instructor**: Operating Systems and Computer Programming labs.